

34 — NAÏVE BAYES CLASSIFIER — IRIS DATASET

Aim

To implement a Gaussian Naïve Bayes classifier using scikit-learn and evaluate its accuracy on the Iris dataset.

Software Required

- Python 3.x
- Jupyter Notebook / Google Colab
- NumPy
- scikit-learn

Algorithm + Procedure

1. Load the Iris dataset.
2. Separate input features and target classes.
3. Split the dataset into training and testing sets.
4. Create a Gaussian Naïve Bayes classifier.
5. Train the classifier.
6. Predict the test data.
7. Calculate accuracy.
8. Generate the confusion matrix.
9. Display the results.

Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix
X, y = load_iris(return_X_y=True)
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.25,
```

```
    random_state=0
)
model = GaussianNB()
model.fit(X_train, y_train)
pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, pred))
print("Confusion Matrix:")
print(confusion_matrix(y_test, pred))
```

Output:

```
Accuracy: 1.0
Confusion Matrix:
[[13  0  0]
 [ 0 16  0]
 [ 0  0  9]]
```